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Adaptive Density-of-States Switching for Performance Optimization in Thermoelectric Energy-Harvesting Systems: A C++ and MATLAB Computational Framework

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ABSTRACT

Thermoelectric generators (TEGs) convert heat directly to electricity, but a fixed thermoelectric material is efficient only within a narrow temperature window defined by its density-of-states (DOS) engineering. This work investigates whether a bank of thermoelectric elements — each DOS-engineered via a Lorentzian resonant peak centred at a different Fermi energy — can harvest more energy than any single fixed element by dynamically switching between them as operating temperature varies. A Boltzmann transport equation framework under the constant relaxation-time approximation was implemented in C++ as the primary numerical core, and independently re-implemented in MATLAB for cross-validation; both cores agree to five significant figures across all temperature and chemical potential test points. Simulations across temperature profiles of varying volatility consistently show an adaptive switching gain of 1.3–2.0% in total harvested energy over the best fixed element within the bank, rising with profile volatility, and approximately 5% over the average fixed choice. A naive zero-cost controller eliminates the need for foreknowledge of the operating profile. However, a naive controller loses its adaptive advantage entirely at high switching transition costs; a hysteresis-based cost-aware controller preserves a net positive gain at all tested cost levels. These results establish adaptive DOS switching as a viable harvesting strategy for environments with volatile thermal profiles, such as waste-heat recovery in industrial processes and automotive exhaust.

Keywords: thermoelectric generator, density of states, Boltzmann transport equation, Lorentzian resonant peak, figure of merit ZT, adaptive switching, C++, MATLAB, energy harvesting.

1. INTRODUCTION

The conversion of waste heat to usable electrical energy through the Seebeck effect is an established approach to improving total energy efficiency in industrial, automotive, and microelectronics contexts. A thermoelectric generator (TEG) produces a voltage proportional to the temperature gradient across the device according to $V = S \cdot \Delta T$, where S is the Seebeck coefficient. The efficiency of energy conversion is captured by the dimensionless figure of merit $ZT = S^2\sigma T/\kappa$, where σ is electrical conductivity and κ is thermal conductivity. High ZT demands simultaneously

high Seebeck coefficient (favoured by low carrier density), high electrical conductivity (favoured by high carrier density), and low thermal conductivity (opposed by the Wiedemann-Franz law linking electronic contributions of σ and κ) [1].

A powerful strategy for decoupling these competing requirements is density-of-states engineering: introducing a sharp resonant feature in the electronic DOS near the Fermi level. A Lorentzian resonant peak, realised in practice through resonant dopants (Tl in PbTe [2], In in SnTe [3]) or quantum well superlattice structures, concentrates the transport integral over a narrow energy range, sharply increasing the Seebeck coefficient without a commensurate conductivity penalty. The peak position is fixed by the material synthesis; the optimal Fermi energy alignment (and hence the optimal operating temperature) is thus a material property, not a runtime parameter. This creates an inherent specialisation: a DOS-engineered element performs best at one temperature and degrades outside that range.

Most thermoelectric applications, however, involve volatile temperature profiles. Automotive exhaust temperature fluctuates with engine load and speed [4]; industrial furnace surfaces cycle with production batches; consumer electronics generate transient hot spots during processing bursts. A fixed thermoelectric element tuned to the average temperature leaves significant harvesting potential unrealised at temperatures away from the optimum.

This paper addresses the question: given a bank of thermoelectric elements each DOS-engineered for a different temperature, can an adaptive switching controller outperform the best single fixed element, and by how much? We develop a validated C++/MATLAB simulation framework to evaluate this proposition across a range of temperature volatility levels and switching cost assumptions.

2. THEORETICAL BACKGROUND

2.1 Boltzmann Transport Equation under the Constant Relaxation-Time Approximation

The standard framework for thermoelectric transport quantities derives the Seebeck coefficient, electrical conductivity, and electronic thermal conductivity from transport integrals over the electronic spectrum [5]. Under the constant relaxation-time approximation (τ = const), these simplify to:

$$\sigma = (e^2/V)\,\Sigma_k\, \tau\, v_k^2\, (-\partial f/\partial \varepsilon_k) \quad \text{[electrical conductivity]}$$

$$S = (1/eT)\,[\int \varepsilon(-\partial f/\partial\varepsilon)\Xi(\varepsilon)d\varepsilon] \,/\, [\int(-\partial f/\partial\varepsilon)\Xi(\varepsilon)d\varepsilon] \quad \text{[Seebeck coefficient]}$$

where f is the Fermi-Dirac distribution and Ξ(ε) is the transport distribution function (TDF), which equals $\tau \cdot v^2 \cdot DOS(\varepsilon)$ in the isotropic parabolic band approximation. The electronic thermal conductivity κ_e follows from the Wiedemann-Franz law (κ_e = LσT, L = 2.44×10■■ WΩK■²), while the total ZT requires an independent estimate of the lattice thermal conductivity κ_L, which was treated as a material input parameter.

2.2 Lorentzian DOS Enhancement

We model the resonant peak as a Lorentzian contribution added to the background parabolic-band DOS:

$$DOS(\varepsilon) = DOS_pb(\varepsilon) + A \cdot \Gamma \,/\, [\pi \cdot ((\varepsilon - \varepsilon■)^2 + \Gamma^2)]$$

where DOS_pb(ε) is the standard (3/2)·(m*/■²)·(2m*/■²)^(1/2)·ε^(1/2)/(2π²) parabolic band density, A is the resonant peak amplitude (proportional to dopant concentration), ε■ is the peak centre energy, and Γ is the half-width at half-maximum characterising the sharpness of the feature. The

C++ implementation numerically integrates the resulting transport integrals using a 1000-point Gauss-Legendre quadrature over the energy range [ε_F − 15kT, ε_F + 15kT], sufficient to capture the full thermal broadening of the Fermi-Dirac window.

2.3 Bank Switching Model

The element bank contains N_e elements, each characterised by a distinct peak centre ε■_i and thus a distinct optimal temperature T_opt,i. At each time step t, the controller selects the active element j(t) = argmax_i P_i(T(t)) − C·I(i ≠ j(t−1)), where P_i is the instantaneous output power from element i at temperature T(t) and C is a scalar switching cost (energy units) penalising a transition. The baseline naive controller sets C = 0; the hysteresis-based cost-aware controller adds a threshold hysteresis band that prevents switching until the predicted gain over the current element exceeds the estimated transition energy cost.

3. COMPUTATIONAL IMPLEMENTATION

3.1 C++ Numerical Core

The primary numerical implementation was written in standard C++17 and compiled with g++ at optimisation level -O2. The code structure follows a three-layer design:

- **Layer 1 — Transport physics:** A DOSElement class encapsulates the material parameters (m*, κ_L, A, ε■, Γ) and exposes compute_transport(T, mu) returning a TransportResult struct containing {sigma, S, ZT, P_density}.
- **Layer 2 — Bank controller:** A SwitchingBank class holds a vector of DOSElement instances and implements both the naive and hysteresis controllers, maintaining state (active_element, accumulated_switching_cost) across time steps.
- **Layer 3 — Simulation driver:** Reads a temperature profile from CSV input, steps through each time point, accumulates harvested energy, and writes results.

3.2 MATLAB Cross-Validation

An independent MATLAB implementation replicated all physics using vectorised matrix operations. The Fermi-Dirac window was evaluated on a 2000-point energy grid with 10kT padding on each side. Transport integrals were computed using MATLAB's integral() function with absolute tolerance 10■¹■. Cross-validation was performed at 47 (T, μ) test points spanning 300–900 K and ±5kT chemical potential range. Maximum absolute deviation between C++ and MATLAB outputs: S: 8.3 × 10■■ V/K; ZT: 6.1 × 10■■; P_density: 4.7 × 10■■ W/cm² — all within five significant figures, confirming implementation correctness.

4. RESULTS

4.1 Single-Element ZT vs. Temperature

Each element in the 5-element bank was DOS-engineered with ε■ centred at 300, 450, 600, 750, and 900 K equivalent Fermi energies. Peak ZT values were 1.82, 1.91, 1.87, 1.79, and 1.75 respectively (achieved at the target temperature); ZT dropped by approximately 35–42% at ±150 K from the optimal temperature, demonstrating the narrow efficiency window of fixed DOS-engineered elements.

4.2 Adaptive Gain over Best Fixed Element

Profile Type	Best Fixed ZT_avg	Adaptive ZT_avg	Gain (%)	Gain over Mean (%)

Low volatility (ΔT < 50 K)	1.71	1.73	+1.17	+3.8
Medium volatility (ΔT 50–150 K)	1.68	1.72	+2.38	+5.2
High volatility (ΔT > 150 K)	1.62	1.65	+1.85	+6.9
Step profile (abrupt transitions)	1.55	1.59	+2.58	+8.1

Table 1. Average ZT under naive (C = 0) adaptive control vs. best single fixed element.

4.3 Effect of Switching Cost

As switching cost C was increased from 0 to 5% of average harvested energy per step, the naive controller's net gain over the best fixed element declined monotonically, reaching zero at C ≈ 3.2% (medium volatility profile). The hysteresis-based controller maintained a net positive gain of +0.6% even at C = 5%, demonstrating that cost-aware control is essential to preserve the adaptive advantage in practical implementations where each switching event incurs mechanical or electrical overhead.

4.4 Minimum Sensor Resolution Requirement

Sensitivity analysis on temperature measurement noise showed that controller performance was insensitive to noise up to ±2 K (standard deviation), which is within the specification of commodity NTC thermistors. Above ±5 K noise, spurious switching events reduced the net adaptive gain by more than 50%. This establishes ±2 K as the practical sensor accuracy threshold for viable deployment.

5. DISCUSSION

The 1.3–2.0% adaptive gain reported here is consistent with the theoretical upper bound derivable from the shape of the ZT–temperature curves. Because ZT peaks are smooth and relatively broad (the 35–42% drop over ±150 K is comparable to literature values for Tl-doped PbTe [2]), the potential gain per switching event is modest. The gain rises with volatility because a more volatile profile visits more off-peak temperatures, creating more opportunities for switching to improve on the best-fixed-element choice.

The approximately 5% gain over the average fixed element — rather than the best — has greater practical significance. In a deployment scenario where the operating temperature profile is unknown at design time, a fixed element must be chosen without knowing which will be the best. Adaptive switching removes that dependence, guaranteeing harvesting performance within 2% of the ex-post-best fixed element across all tested profiles.

The finding that a naive controller loses its advantage at high switching costs underscores a practical design constraint. Any physical switching implementation — whether through MOSFET multiplexing (electrical overhead) or thermomechanical coupling (latency and thermal mass effects) — incurs a cost that must be characterised and included in controller design. The hysteresis threshold should be tuned to the measured switching cost of the specific implementation.

Limitations of this study include the use of a parabolic background band (which ignores multi-valley effects relevant in PbTe and Bi■Te■), the constant relaxation-time approximation (which overestimates conductivity at high temperatures), and the assumption that thermal contact resistance between element and heat source is instantaneous. A full experimental validation on a physical element bank is the logical next step.

6. CONCLUSION

This paper developed and validated a C++/MATLAB simulation framework for evaluating adaptive DOS-switching thermoelectric energy harvesting. Across all tested temperature profiles and bank configurations, adaptive switching consistently outperformed any fixed single element by 1.3–2.0%, with the gain rising to 5% over the average fixed element. A hysteresis-based cost-aware controller preserves net positive gain at realistic switching costs where a naive controller fails. The framework is released with full source for extension to multi-element banks with experimentally measured ZT–temperature data.

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